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May 9, 2024

Docket Nos.: 50-321

NL-24-0166

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant
LER 2024-002-00
Unit 1 Manual Reactor Trip Due to Loss of Feedwater

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73 (a)(2)(iv)(A), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins, at 912.453.2342.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "M. Busch".

Matt Busch
Vice President – Hatch

MSB / JMH

Enclosure: LER 2024-002-00

Cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Enclosure to NL-24-0166
LER 2024-002-00

**Edwin I. Hatch Nuclear Plant Unit 1
Licensee Event Report 2024-002-00**

Manual Trip Due to Loss of Feedwater

Enclosure

LER 2024-002-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Edwin I. Hatch Nuclear Plant Unit 1



050



052

2. Docket Number

321

3. Page

1 OF 2

4. Title

Unit 1 Manual Reactor Trip Due to Loss of Feedwater

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	11	2024	2024	- 002 -	00	05	09	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode

MODE 1

10. Power Level

035

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Jimmv Collins - Licensing Manager

Phone Number (Include area code)

912-453-2342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

15. Expected Submission Date

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 1338 EDT on 3/11/2024, while Unit 1 was in MODE 1 at 35% power and performing power ascension activities, the reactor was manually tripped due to reactor water level decreasing following the trip of the 'A' Reactor Feed Pump (RFP) on low suction pressure. Due to the power level at the time, the 'B' RFP had not been placed in service. Closure of Containment Isolation Valves (CIV's) in multiple systems and actuation of High Pressure Coolant Injection (HPCI) and Reactor Core Isolation Cooling (RCIC) pumps occurred as a result of reaching the respective actuation setpoints on reactor water level as designed. The 'B' RFP was placed in service to control water level at 1340 EDT on 3/11/2024.

The cause of the 'A' RFP trip was the use of the incorrect procedure section for isolation of a condensate pump while the main condenser was under vacuum. This allowed air leakage into the system which resulted in the low suction pressure trip of the 'A' RFP. The Condensate & Feedwater System procedure has been revised to clearly direct the performer to the appropriate section.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME Edwin I. Hatch Nuclear Plant Unit 1	☒ 050	2. DOCKET NUMBER 321	3. LER NUMBER				
	☐ 052		<table border="1"><tr><td>YEAR</td><td>SEQUENTIAL NUMBER</td><td>REV NO.</td></tr><tr><td>2024</td><td>002</td><td>00</td></tr></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2024
YEAR	SEQUENTIAL NUMBER	REV NO.					
2024	002	00					

NARRATIVE**EVENT DESCRIPTION**

At 1338 EDT on 3/11/2024, while Unit 1 was in MODE 1 at 35% power and performing power ascension activities, the reactor was manually tripped due to reactor water level decreasing following the trip of the 'A' Reactor Feed Pump (RFP)(EIIS Code: P) on low suction pressure. Due to the power level at the time, the 'B' RFP had not been placed in service. Closure of Containment Isolation Valves (CIV's)(EIIS Code: ISV) in multiple systems and actuation of High Pressure Coolant Injection (HPCI)(EIIS Code: BJ) and Reactor Core Isolation Cooling (RCIC)(EIIS Code: BN) pumps occurred as a result of reaching the respective actuation setpoints on reactor water level as designed. The 'B' RFP was placed in service to control water level at 1340 EDT on 3/11/2024. Once the 'B' RFP was placed in service, HPCI and RCIC were secured at 1340 EDT and 1342 EDT respectively.

EVENT CAUSE ANALYSIS

Based on causal analysis, the cause of the 'A' RFP trip was the use of the incorrect procedure section for isolation of a condensate pump while the main condenser was under vacuum. This allowed air leakage into the system which resulted in the low suction pressure trip of the 'A' RFP.

SAFETY ASSESSMENT AND REPORTABILITY

There were no actual safety consequences as a result of this event. All safety related systems functioned as designed during this event. The operating crew responded correctly to the event. The applicable abnormal/emergency operating procedures were entered. The event was within the analysis of the Updated Final Safety Analysis Report (UFSAR) Chapter 15.

This event is reportable per 10 CFR 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph (a)(2)(iv)(B) of this section. Systems that actuated during this event that are listed in (a)(2)(iv)(B) include; the Reactor Protection System (RPS)(EIIS Code: IG), general Containment Isolation Valves (CIV) in more than one system, ECCS including the HPCI system, and the RCIC system.

CORRECTIVE ACTIONS

The Condensate & Feedwater System procedure has been revised to clearly direct the performer to the appropriate section for isolating a condensate pump when the main condenser is under vacuum.

PREVIOUS SIMILAR EVENTS

On 09/29/2000 at 1357 EDT, Unit 1 was in the Run mode at a power level of approximately 1520 CMWT (55 percent rated thermal power). At that time, the reactor was scrammed manually following a trip of the operating Reactor Feedwater Pump (RFP) and a decrease in water level. Personnel inserted the scram in anticipation of an automatic scram on low water level. A RFP was placed into service to maintain level. This event was caused by component failure. During the implementation of a clearance to isolate a condensate pump, air was introduced into the condensate pump suction header through a leaking pump suction isolation valve. The presence of air led to the trip of the operating RFP on low suction pressure. LER 2000-011-00.